

Academic Year	Module	Assessment Number	Assessment Type
2024	5CS037 Concepts and Technologies of AI	1	Report

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Title: Analysis of the World Happiness Report: Exploring South Asia and Middle East Perspectives.

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Introduction:

World Happiness Report is yearly published by the United Nations Sustainable Development Solutions Network, that labels countries based on the happiness and well-being of their citizens. There are plenty of factors which comes in consideration to rank countries on happiness report such as: Literacy, Life span, Health, Quality of life, death rate, income, average wage salary, employment rate, and so on.

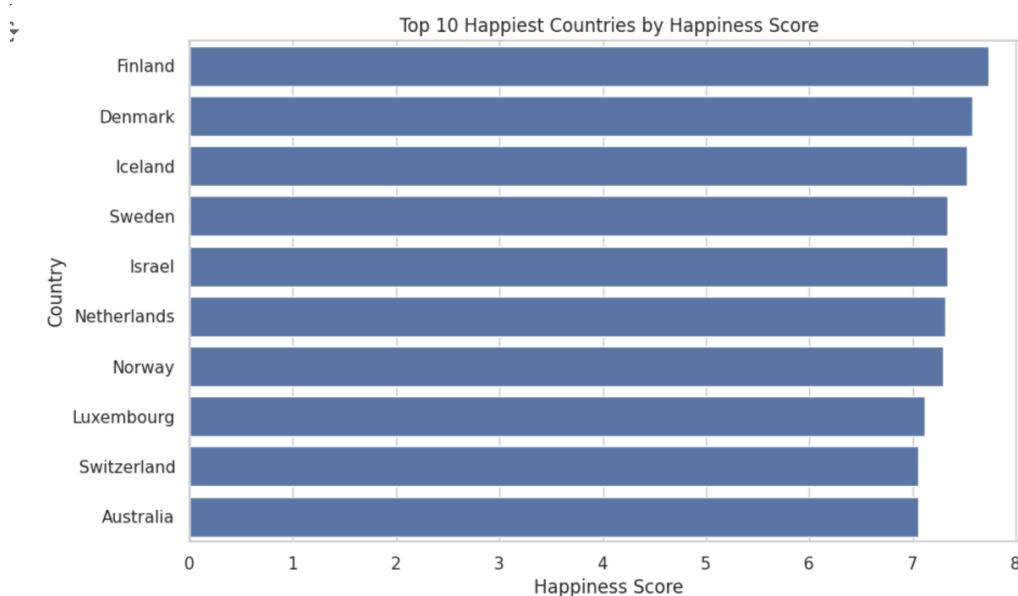
World happiness report is very important for policy makers since it gives back a reality check on what factors they should work in future to make a significant change. Also, it is very important since it lets everyone know which country has best environment for all kind of people. Simultaneously, encourage governments to base their policy on evidence: By doing so, we hope that they can implement policies which contribute to a much higher level of life satisfaction and wellbeing for all.

The major intend of this report is to write about world happiness report and it importance ,simultaneously to describe what we learned while doing assignment 1 and how we used different python libraries.To keep it short in introduction ,we firstly worked on data exploration which includes variables This section involves identifying trends, patterns, and potential correlations in the data that contribute to happiness rankings with showing its plot diagrams and scatter diagrams. In second task we focused on analyzing the happiness scores

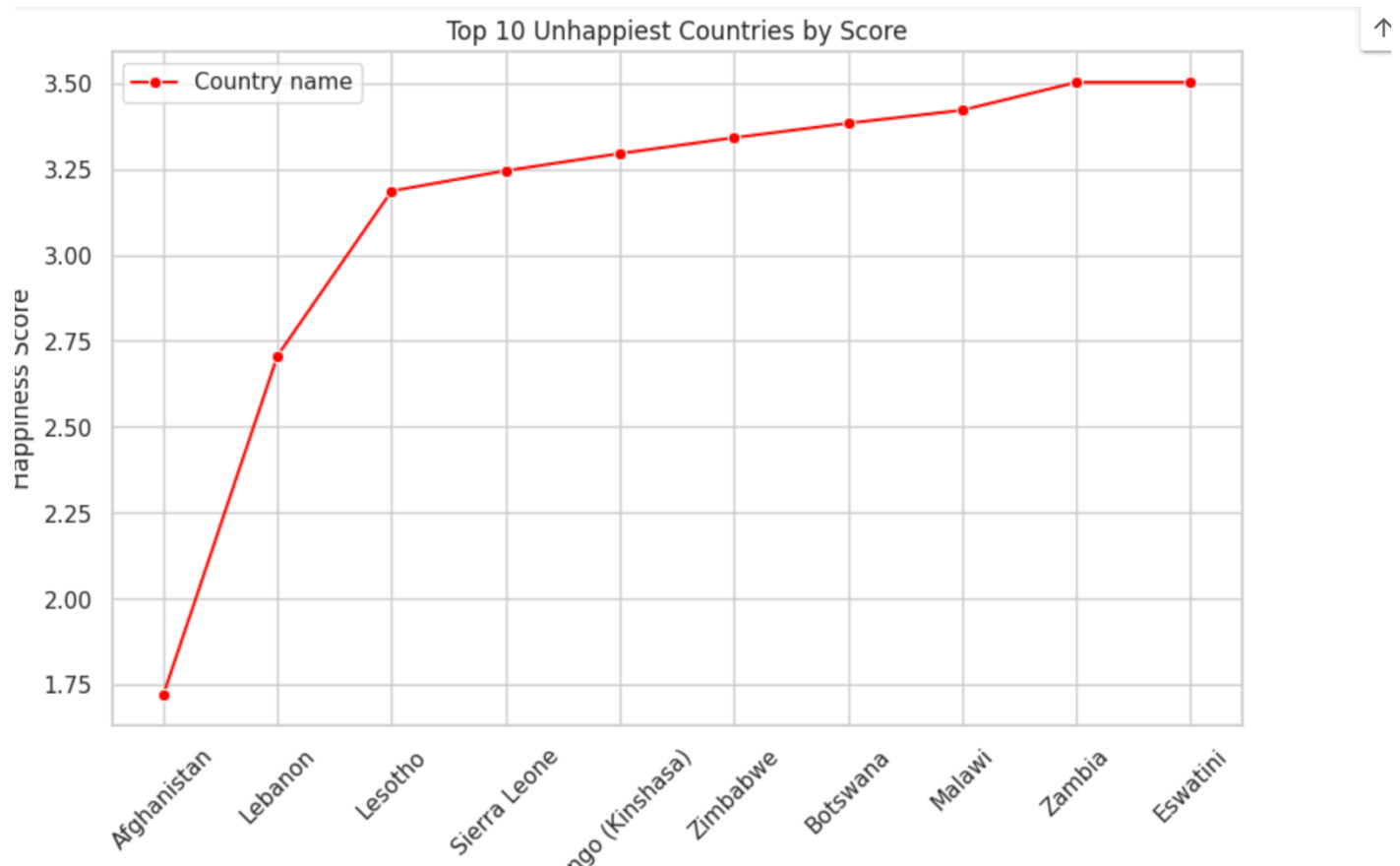
for South Asian countries which includes comparing many reason like GDP, and the happiness scores of these countries. The major focus here was to understand regional disparities and factors that uniquely affect the well-being of South Asian nations. And the final task was to put them together and know about them between the happiness scores of South Asian countries with those of the Middle Eastern nations. In this section I was looking for differences and similar factors influencing happiness, such as economic performance, social support, and government policies in both regions.

Problem 1:

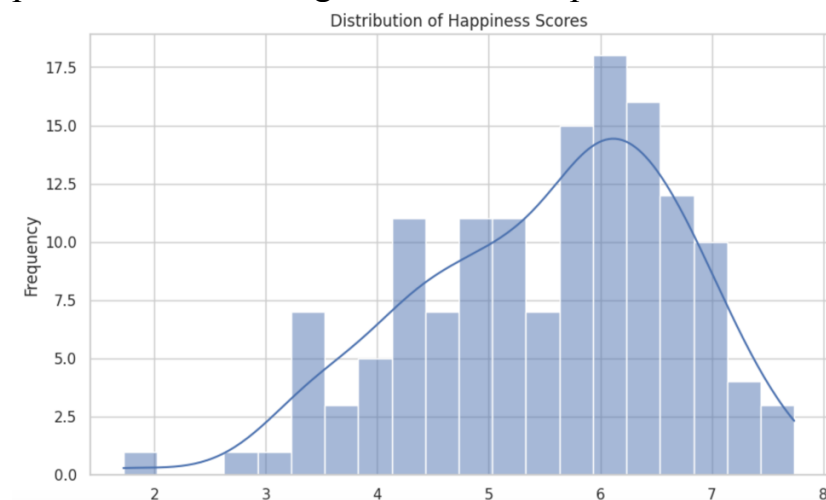
Problem 1 was about Data Exploration and Understanding where we were supposed to show dataset after that present first 10 rows and find the number of columns and rows and then put entire columns and respective data types, all these were in data overview section and in the next section called basic statistics we calculated to find the mean, median, and standard deviation using its formula for the column. After that we checked whether other missing values exist or not in data. and filtered the dataset to present countries with a Score greater than 7.5 as question has asked then we sorted the dataset by GDP per Capita in descending order and displayed top 10 rows. Simultaneously, now we created a column called Happiness_category which was asked to categories countries in 3 categories called high, medium, low. In data visualization, we plotted the top 10 happiest countries by Score using different codes which are in my end link of this report:

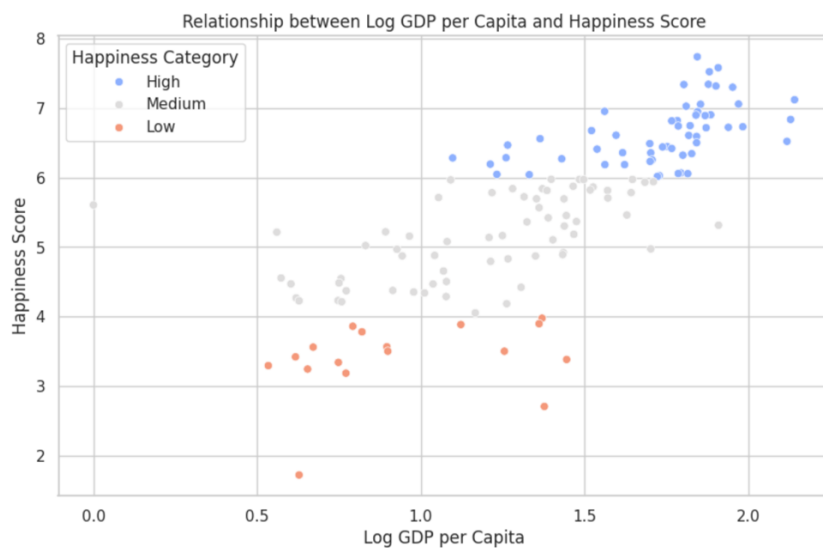


And in second part we plotted the top 10 unhappiest countries by Score using a Line chart as shown below:



Simultaneously, in 3rd and 4th part we plotted histogram and scatter plot for score column and to find between GDP per Capita and Score to visualize their relationship. Below are the picture of both histogram and scatter plot:





Problem 2:

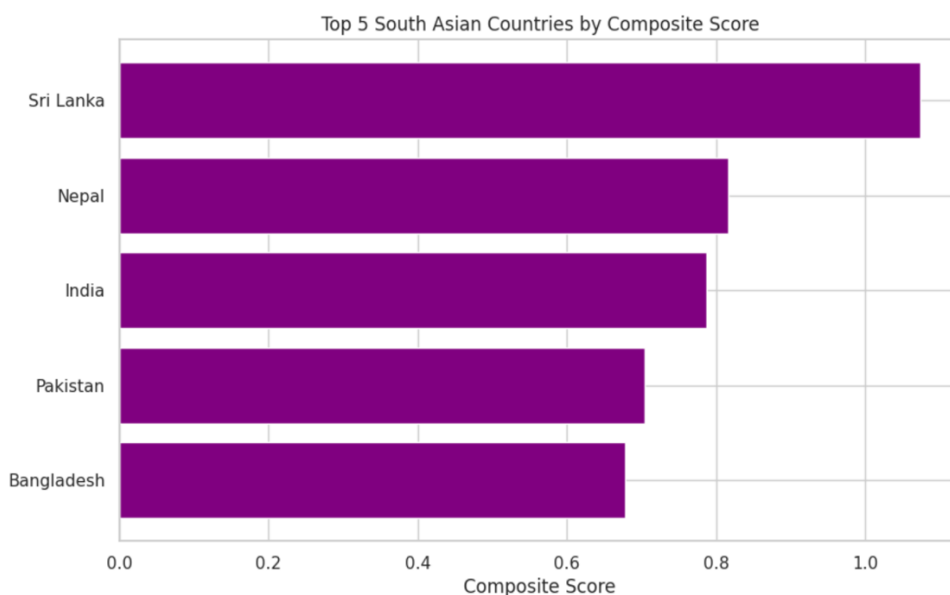
Here we were setting up for South Asian dataset which will be easy in other remaining codes.

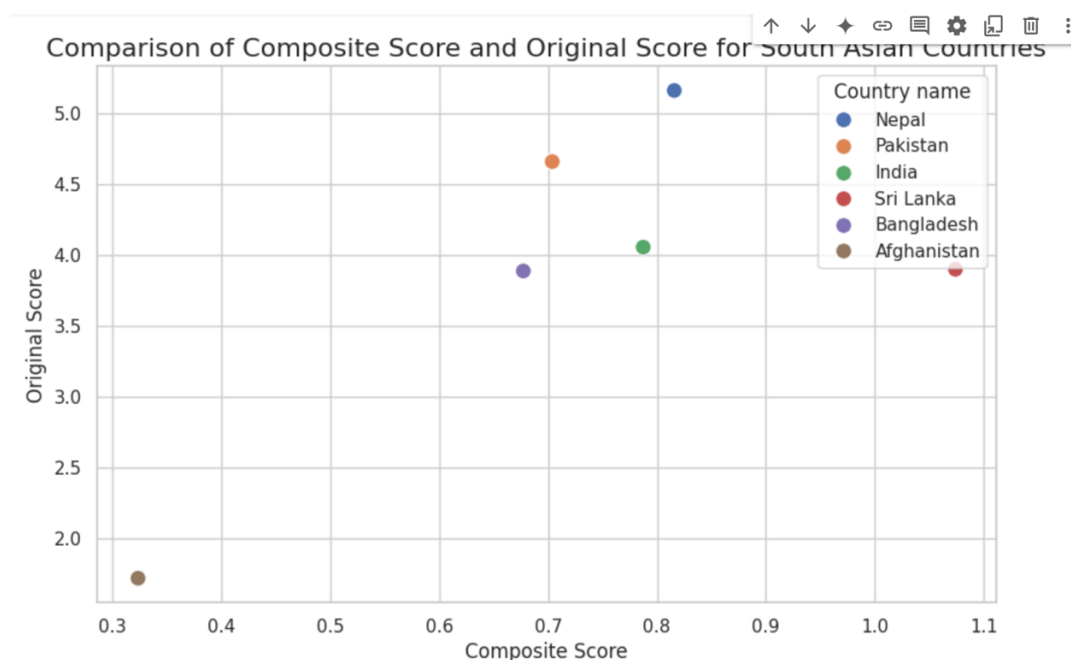
we filtered out the matching dataset and saved filtered data frame as separate csv file named (**south_asia_countries.csv**) Now, Using the South_Asia_countries DataFrame we created a new column which combines the following metrics as question then we worked on ranking south asian countries depending upon composite score in descending order whose output came as such with code

```
# Rank the countries based on Composite Score in descending order Qno 19
SouthAsia['Rank'] = SouthAsia['Composite Score'].rank(ascending=False, method='min')

# Display the first few rows to check the ranking
print(SouthAsia[['Country name', 'Composite Score', 'Rank']].head())
```

	Country name	Composite Score	Rank
0	Nepal	0.8159	2.0
1	Pakistan	0.7039	4.0
2	India	0.7874	3.0
3	Sri Lanka	1.0739	1.0
4	Bangladesh	0.6774	5.0



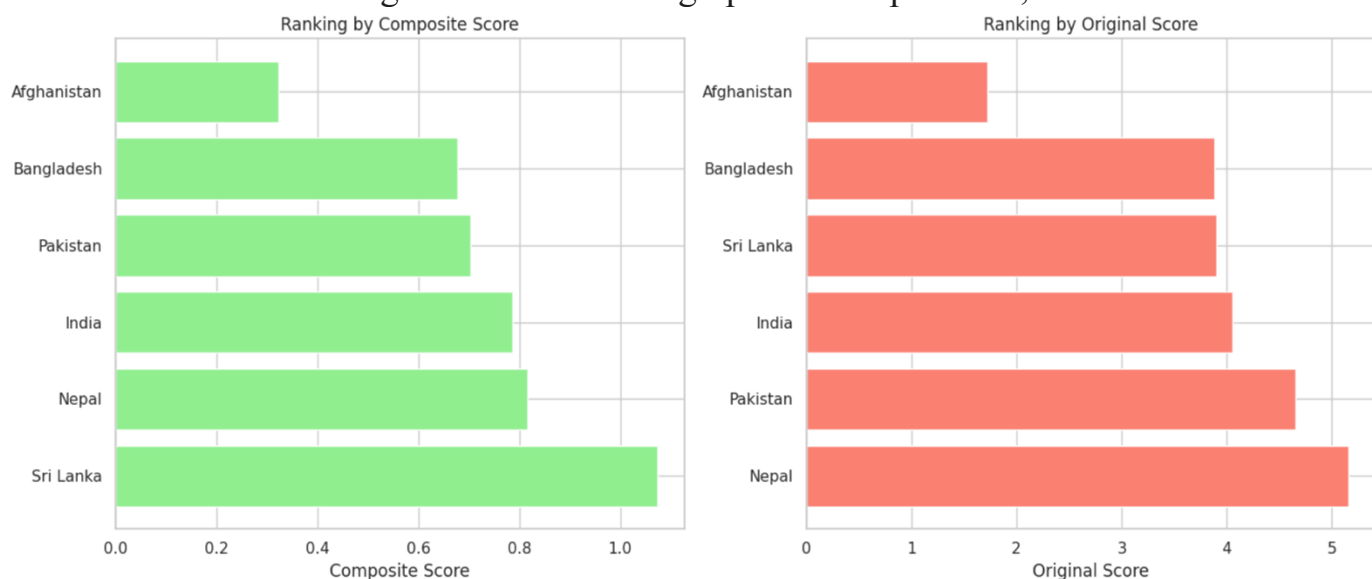


With respect to the plot and data, the South Asian countries' raw score and composite score comparisons are as indicated below:

Sri Lanka, with its robust social and economic parameters stands first in the composite score followed by Nepal and India. Afghanistan is at the bottom.

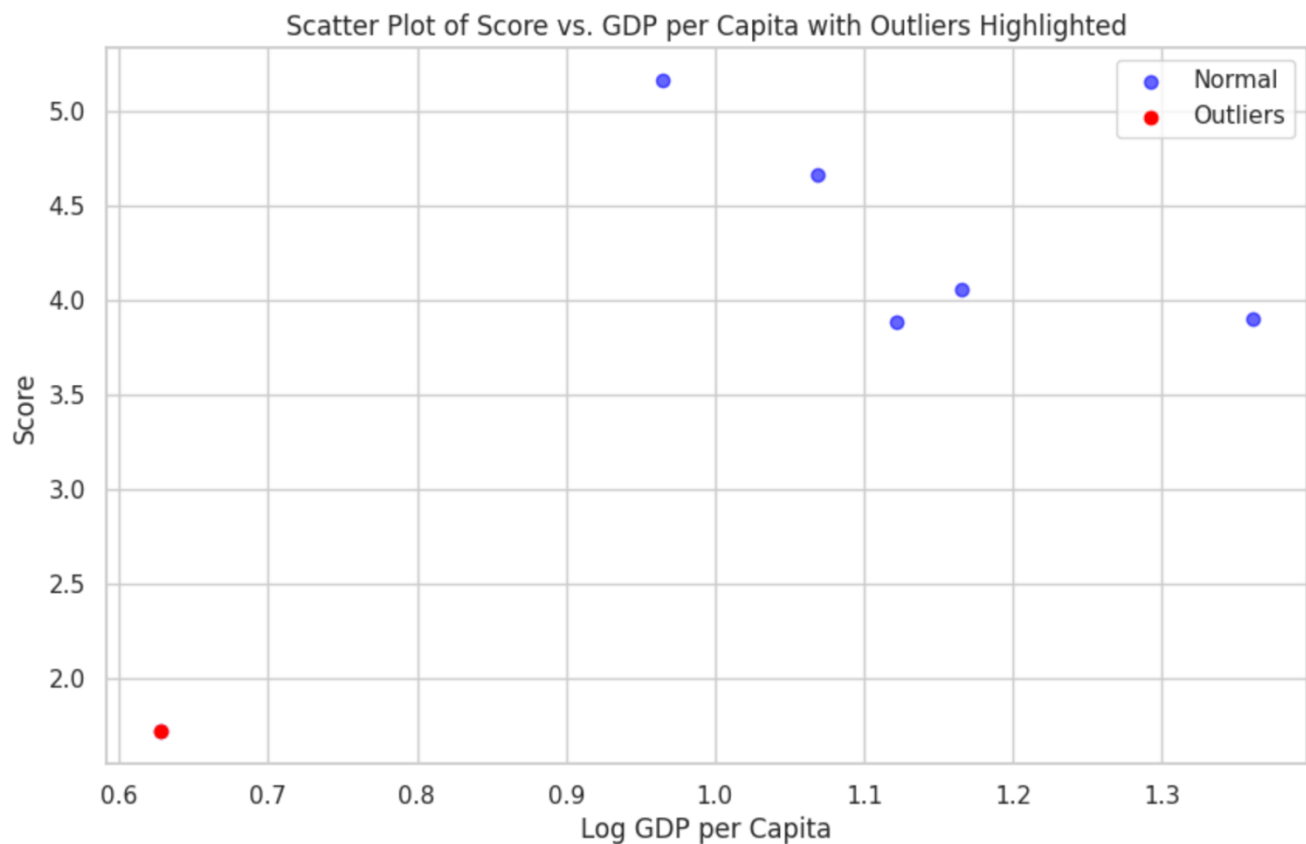
Nepal has the best happiness score (or life satisfaction) and Afghanistan the worst (great difficulties).

Key Takeaway: Balanced metrics, similar as between Nepal to India, become necessity for overall ultimate well-being. I do have another graph for comparison ,



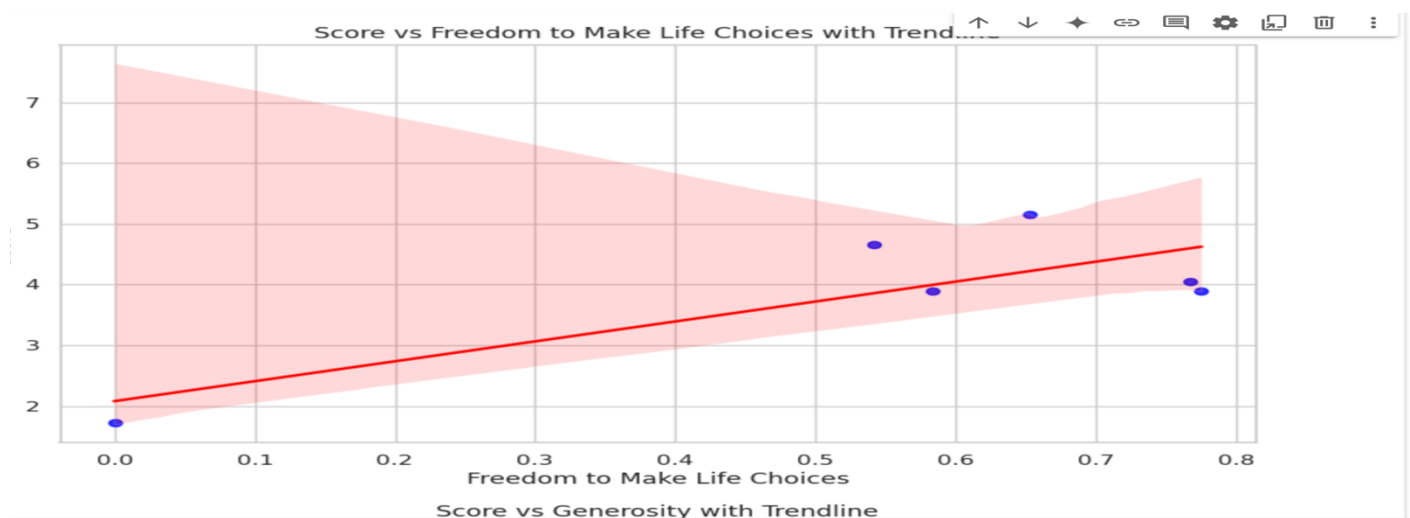
In task 3

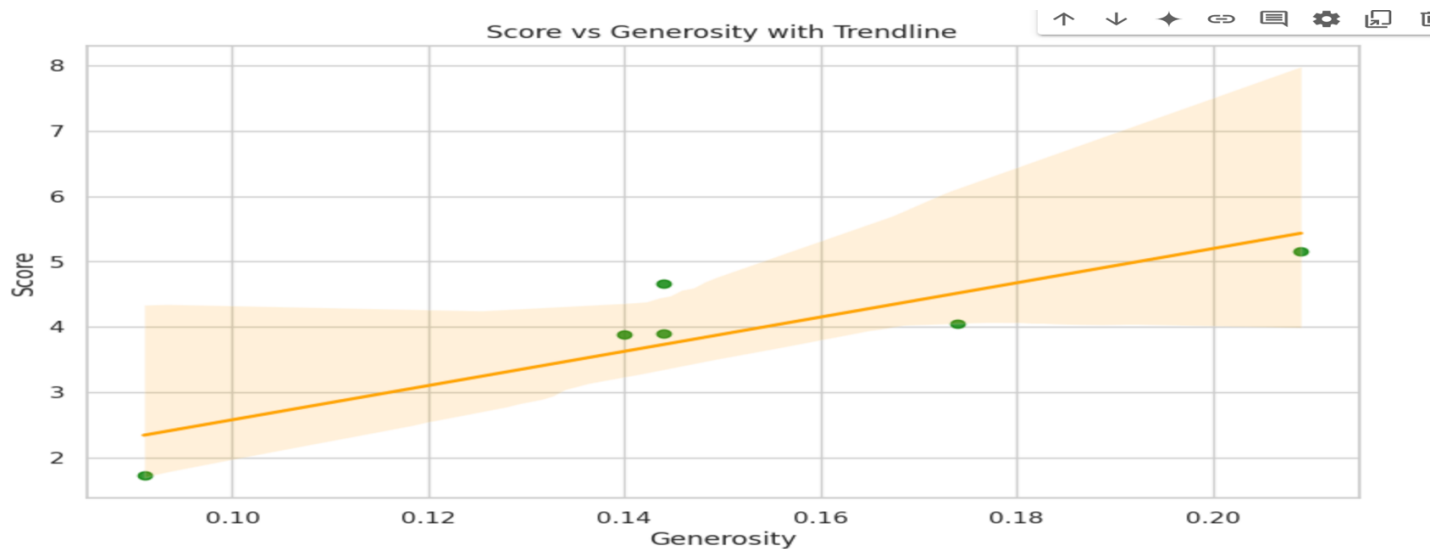
The Interquartile range rule (IQR rule) One of the common methods for defining outliers is to use the $1.5 \times \text{IQR}$ rule (Interquartile Range Rules)



Q4 of task 3:

In comparison to other nations in South Asia, Afghanistan is well behind in terms of happiness score which also brings down the overall average of the region. In this regard, it is necessary to take measures in the country with a view to elevating general wellbeing and GDP per person income. The answer of task 4 (1) are 0.8005 of Pearson correlation between Score and Freedom to Make Life Choices and 0.874512371253192 of Pearson correlation between Score and Generosity. Below is the scatter plot of trendlines for these metrics against the Score





After that we have added a column GDP per Capita and the Score for here is its output :

Step 4: Create the 'GDP-Score Gap' column Qno 28

```
SouthAsia['GDP-Score Gap'] = SouthAsia['Log GDP per capita'] - SouthAsia['score']
```

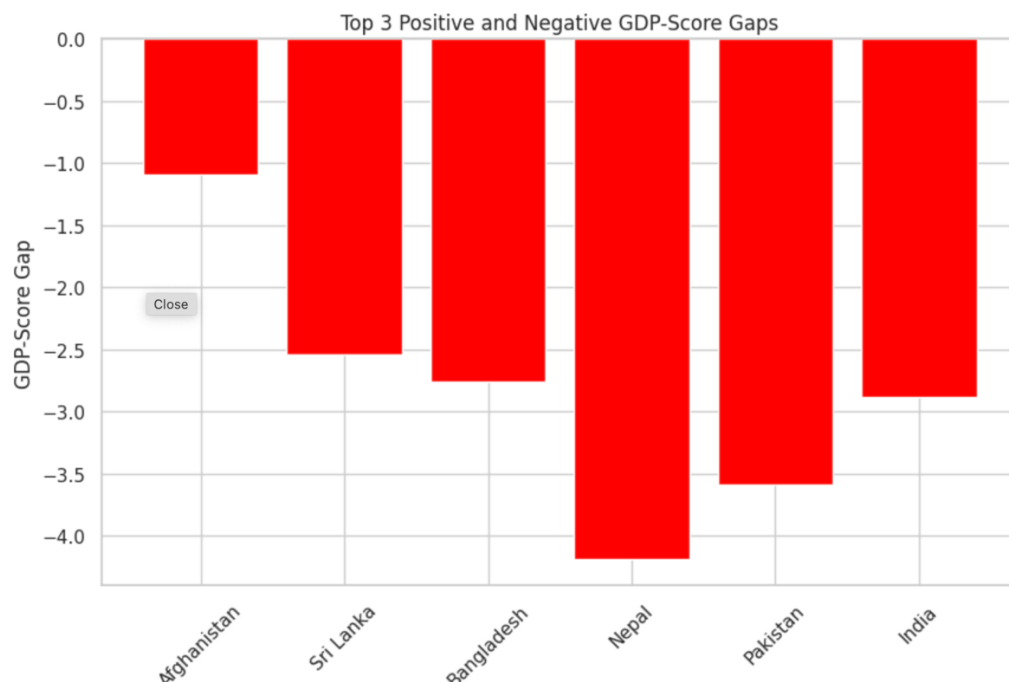
Step 5: Check the result

```
print(SouthAsia[['Country name', 'Log GDP per capita', 'score', 'GDP-Score Gap']].head())
SouthAsia.head()
```

GDP per capita	score	GDP-Score Gap
0.965	5.158	-4.193
1.069	4.657	-3.588
1.166	4.054	-2.888
1.361	3.898	-2.537
1.122	3.886	-2.764

```
:
```

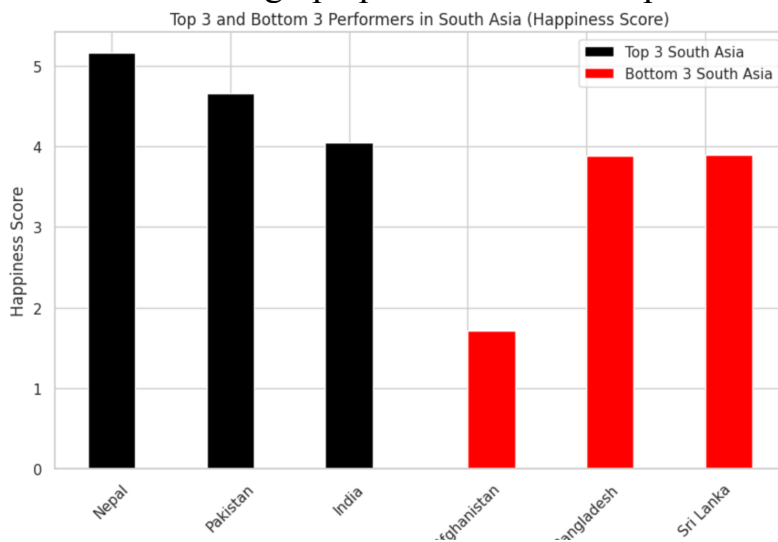
	Country name	GDP-Score Gap	Rank by Gap Ascending	Rank by Gap Descending
0	Nepal	-4.193	1.0	6.0
1	Pakistan	-3.588	2.0	5.0
2	India	-2.888	3.0	4.0
3	Sri Lanka	-2.537	5.0	2.0
4	Bangladesh	-2.764	4.0	3.0

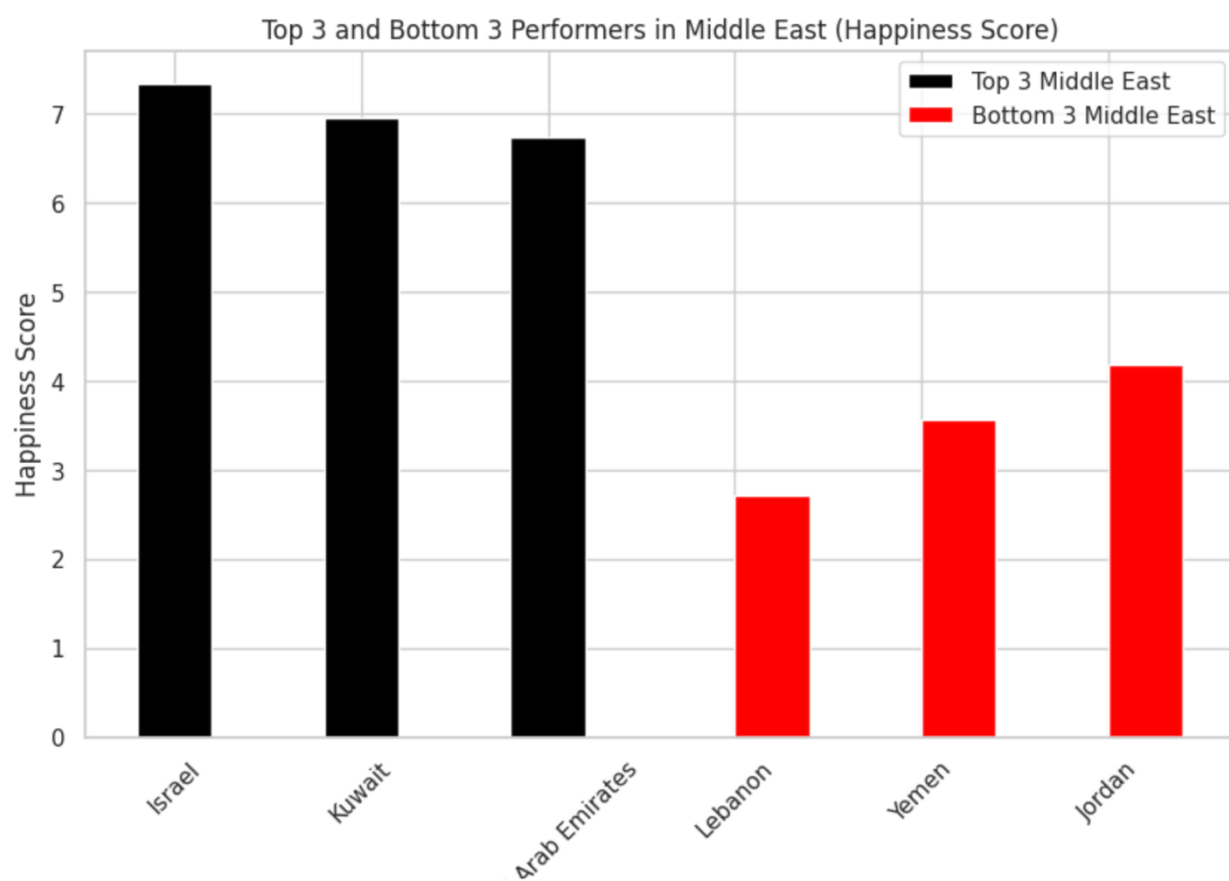


Ans) Happiness gaps in south asia are due to economic inequality, social support difference emphasizing the need for balanced growth and effective governance.

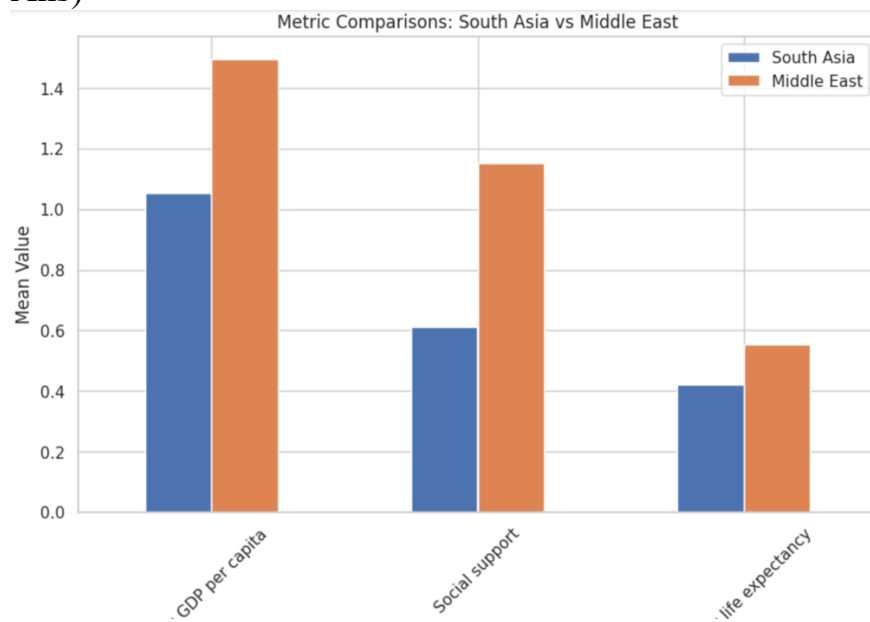
Problem 3:

Below are all the graph plots and charts of problem 3 tasks :

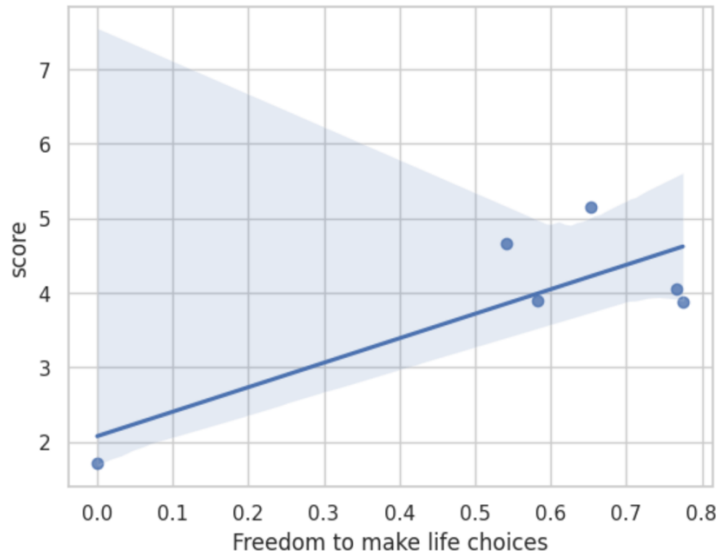




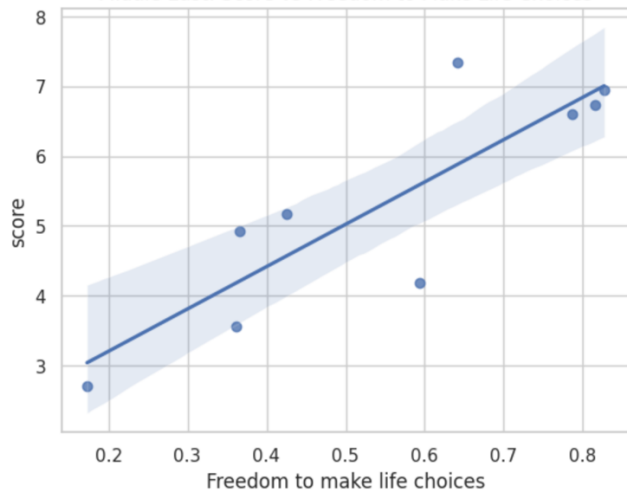
Ans)



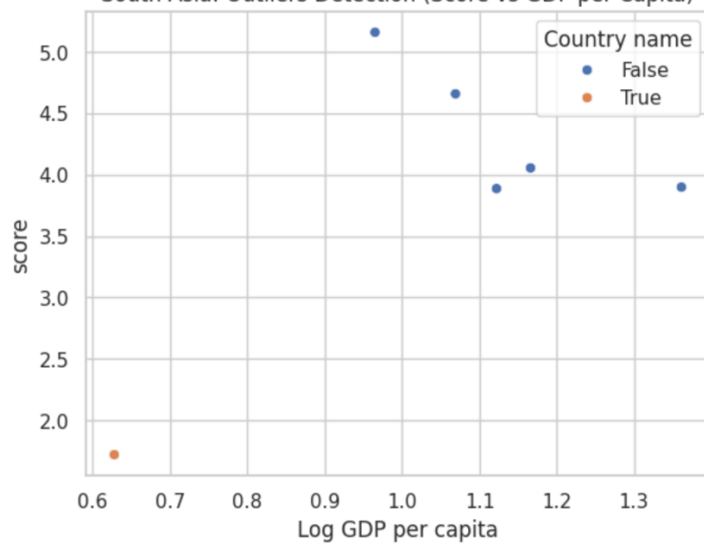
South Asia: Score vs Freedom to Make Life Choices

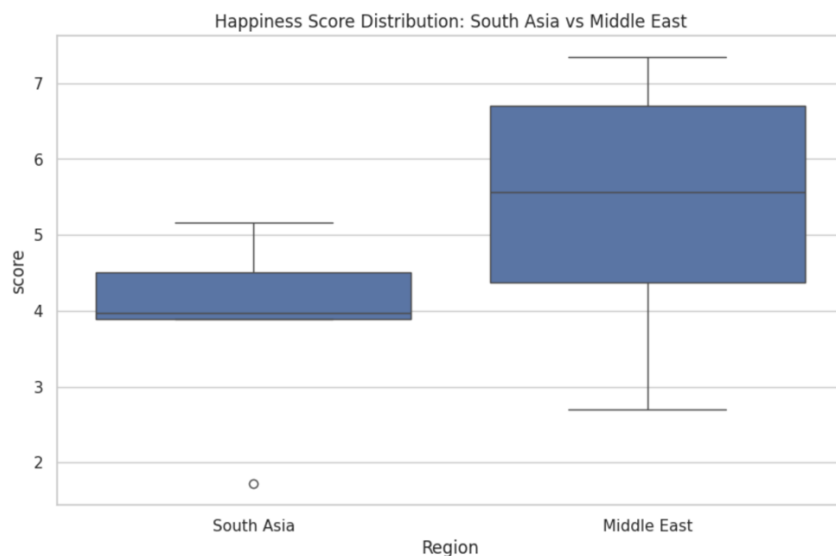
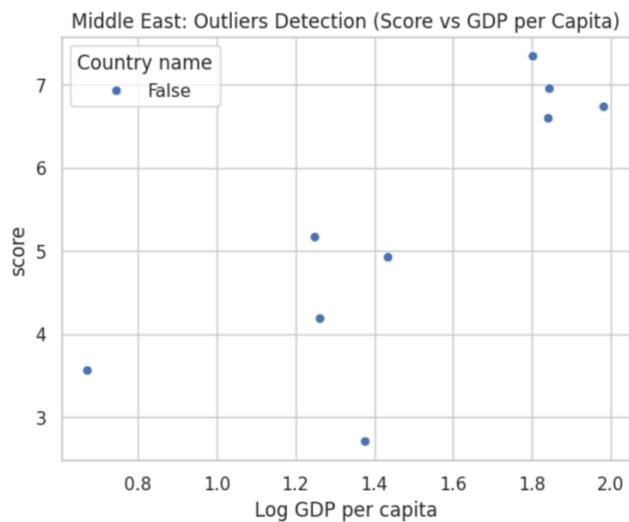


Middle East: Score vs Freedom to Make Life Choices



South Asia: Outliers Detection (Score vs GDP per Capita)





Conclusion: After the completion of assignment 1 I was able to learn about dataset and how to plot different type of graphs and most importantly I was able to learn and use libraries such as numpy, pandas. Thank you module leader : Siman sir and other respected teachers for providing us with this opportunity to challenge ourselves.

Github link:

[https://github.com/Sandyon1top/Workshop1.AI/blob/main/2408003_SANDIP_Dhakar%20\(1\).ipynb](https://github.com/Sandyon1top/Workshop1.AI/blob/main/2408003_SANDIP_Dhakar%20(1).ipynb)